Technical Report for ARGUS: Beyond Counts, Rate-Based Sketching for Persistence-Based Item Lookup

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This report accompanies the conference paper of the same name and contains only the material the paper has no room for: the statements Section V of the paper defers to this report (Section I below), the proofs of those and of the statements the paper gives (Section II), the two bucket-layout controls (Sections III and IV), and the figures omitted for space (Section V). Theorem and section numbers of the form *V.x* refer to the conference paper, as do the figure and section numbers marked *of the paper*; numbering here is otherwise independent. The report is self-contained for the proofs, but the statements being proved are given in the paper and are not repeated.

I. Additional Statements

These complement §V of the paper: the statements it gives in prose or leaves to a pointer, the collision terms behind the false-negative guarantees and their sample-complexity forms, the definition of the count-only rule ARGUS-V, and what the bounds become under weaker dependence assumptions.

Theorem 1 (FN bound, full tracking)**.** *Let $p(u) \geq \alpha+\Delta$ with $\Delta \in (0, 1-\alpha]$. Then $\Pr[\{u \notin R_{\text{PI}}\} \cap \mathcal{F}_T(u)] \leq \Pr[V_t(u)/T < \alpha] \leq e^{-2T\Delta^2}$, and for* ARGUS-$\star$ *with $\psi_\star(u) = 1$, $\Pr[\{u \notin R_\star\} \cap \{\hat{\psi}_\star(u) = 1\} \cap \mathcal{F}_T(u)] \leq e^{-2T\Delta^2}$.*

Theorem 2 (Estimation error)**.** *Assume uniform hashing, the working model, and the standard convention that a colliding arrival is independent of the occupant's adverse-verdict event. Let u be an item with $p(u) \geq \alpha + \Delta$, $\Delta \geq 0$ (with $1/(2\Delta^2) := \infty$ at $\Delta = 0$), let $\|p\|_1 := \sum_{u'} p(u')$ and $B_\Delta := \min\{T\delta/2,\ 1/(2\Delta^2)\}$. Then $\widehat{V}(u) \leq V_t(u)$ deterministically and*

$$\Pr[\{\widehat{V}(u) < V_t(u)\} \cap \mathcal{D}(u)] \leq \frac{\|p\|_1 - p(u)}{w}\big(B_\Delta + G_\alpha\big). \quad (1)$$

Corollary 3 (FN bound under eviction, base ARGUS)**.** *Under the hypotheses of Theorem 2 with $\Delta > 0$,*

$$\Pr[\{u \notin R_{\text{PI}}\} \cap \mathcal{D}(u)] \leq \underbrace{e^{-2T\Delta^2}}_{\textit{Hoeffding}} + \underbrace{\frac{\|p\|_1 - p(u)}{w}\Big(\min\Big\{\frac{T\delta}{2},\ \frac{1}{2\Delta^2}\Big\} + G_\alpha\Big)}_{\textit{collision}}. \quad (2)$$

What the collision term would require. The collision term of (2) carries the factor B_Δ that a count-based rule lacks, since a confidence-bounded rule acts on a collision with probability at most $\delta/2$ (§V-E of the paper). Summed over an episode it nonetheless exceeds one unless a row is far wider than a cache-resident budget affords, 1 to 2.6×10^5 buckets against the 320 to 1,920 of 10 to 60 KB. The assembled bounds are therefore asymptotic in w, and at deployable sizes the Hoeffding term is the operative one.

Corollary 4 (FN bound under eviction, variants)**.** *Let $\star \in \{\text{PS}, \text{PH}, \text{PP}\}$, $p(u) \geq \alpha + \Delta$ with $\Delta > 0$, $\psi_\star(u) = 1$, under the hypotheses of Theorem 2 (and, for* PP*, the admission condition stated in Remark 5, so $q_{\text{PP}}(\tau) = 0$). Let $q_\star(\tau) := \sup_u \Pr[\{\hat{\psi}_\star(u) \neq \psi_\star(u)\} \cap \mathcal{F}_\tau(u)]$ be the shortcut's per-collision misjudgement probability, $\hat{\psi}_\star$ read at tracked length τ against the query-time $\psi_\star$. Then*

$$\Pr[\{u \notin R_\star\} \cap \mathcal{D}(u)] \leq e^{-2T\Delta^2} + \frac{\|p\|_1 - p(u)}{w}\Big(\min\Big\{\frac{T\delta}{2},\ \frac{1}{2\Delta^2}\Big\} + \sum_{\tau \leq T} q_\star(\tau) + G_\alpha\Big). \quad (3)$$

Remark 5 ($q_\star$ calibration). Let $\bar{q}_\star$ be as in Theorem V.5 and $q_\star(\tau)$ as in Corollary 4. A bucket never returns to the empty state ($V \geq 1$ after an admission), so a tracked item is matched by every one of its subsequent arrivals and $\widehat{C}$ counts all of them: at query time on $\mathcal{F}_T(u) \cap \mathcal{D}(u)$ this gives $\widehat{C} = C_t$, $\widehat{V} = V_t$ and $\widehat{\pi} = \pi_t$, hence $\hat{\psi}_\star = \psi_\star$, so that $\bar{q}_\star$ is at most $\sup_u \Pr[\mathcal{F}_T(u)^c \cap \mathcal{D}(u)]$ plus $\Pr[\mathcal{D}(u)^c]$, the first term bounded by Theorem 2. Per collision: for PP, $\hat{\psi}_{\text{PP}} = 1 \wedge \psi_{\text{PP}} = 0$ needs $\widehat{\pi} \geq \pi_{\min} > \pi_t$, impossible by Lemma V.1, and if u's first arrival already carries a tag $\geq \pi_{\min}$ the converse is impossible on $\mathcal{F}_\tau(u)$, so $q_{\text{PP}}(\tau) = 0$ there (after a re-admission $\widehat{\pi}$ restarts from the re-admission tag). For

PS/PH, $\hat{\psi}_\star$ compares $\widehat{C}/\widehat{V}$ against $b \in \{\theta, \kappa\}$; on $\mathcal{F}_\tau(u)$ that ratio covers u's first τ windows and estimates the population mean multiplicity ρ_t from $\widehat{V} \approx p(u)\tau$ active windows; for per-window multiplicities that are i.i.d. with mean ρ_t and bounded support, Hoeffding gives $q_\star(\tau) = O(e^{-c\,|\rho_t - b|^2\, p(u)\tau})$, a heuristic rather than a bound because that hypothesis is additional to the working model. The predicate is thus misjudged only near the sparsity or heaviness boundary, and less often the longer an item is tracked. Off $\mathcal{F}$ the unit decay biases the ratio upward, $(\widehat{C}-k)/(\widehat{V}-k) \geq \widehat{C}/\widehat{V}$ after k decays since $\widehat{C} \geq \widehat{V}$, pushing PS toward misses and PH toward false reports.

Corollary 6 (Sample complexity, base ARGUS)**.** *Under the hypotheses of Theorem 1, $T \geq \ln(1/\eta)/(2\Delta^2)$ implies $\Pr[\{u \notin R_{\text{PI}}\} \cap \mathcal{F}_T(u)] \leq \eta$, and the same for* ARGUS-$\star$ *with $\{\hat{\psi}_\star(u) = 1\}$ intersected in.*

Corollary 7 (Sample complexity, variants)**.** *Under the hypotheses of Corollary 4, for any target $\eta \in (0,1)$, if*

$$T \geq \frac{\ln(2/\eta)}{2\Delta^2},$$
$$w \geq \frac{2(\|p\|_1 - p(u))}{\eta}\Big(\min\Big\{\frac{T\delta}{2}, \frac{1}{2\Delta^2}\Big\} + \sum_{\tau \leq T} q_\star(\tau) + G_\alpha\Big), \tag{4}$$

then $\Pr[\{u \notin R_\star\} \cap \mathcal{D}(u)] \leq \eta$.

Corollary 8 (Heterogeneous lengths)**.** *Let two of the d candidates at a collision, u and u', satisfy $T(u) = T_1 < T_2 = T(u')$, $V(u) = V(u') = V$ and $\text{status}(u) = \text{status}(u') = 0$, and let every other candidate have a count strictly larger than V and an unclipped score larger than $\max\{V/T_1 - \varepsilon(T_1),\ V/T_2 - \varepsilon(T_2)\}$. Then:*

1) ARGUS-*V penalizes the short-history item u with probability exactly $1/2$, irrespective of α, δ, T_1, T_2 (with row-order tie-breaking, in expectation over the uniform placement of u among the d rows, §V-E of the paper).*
2) ARGUS-*LCB penalizes u only if $V/T_1 < \alpha$, either via $\text{UCB}(u) < \alpha$ or, when $V = 1$ and $T_1 > 1/\alpha$, via the override; in particular never when $V/T_1 \geq \alpha$ (e.g., an item present in every window since admission, Item A of Figure 1 of the paper). When u' is the victim it is penalized, for $V \geq 2$, if and only if $V/T_2 + \varepsilon(T_2) < \alpha$, and for $V = 1$ if and only if $1/T_2 < \alpha$, i.e., exactly when its long history rules out persistence (Item B).*

The idealized count-only rule. ARGUS-V of §V-E of the paper differs from ARGUS only in Stages 2 and 4:

$$\text{ARGUS-LCB}: v^* := \arg\min_i(\hat{p}_i - \varepsilon(T_i)),\ \text{then Stage 4};$$
$$\text{ARGUS-V}: v^* := \arg\min_i V_i \text{ (ties by row order)},$$
$$\text{replace if } V(v^*) = 1,\ \text{decrement otherwise.}$$

The comparison is between decision rules, not a bound on any prior system. Of the two differences the pick is the smaller: choosing the victim by smallest count rather than by smallest lower bound moves F1 by at most 0.8% on any stream at any budget, and treating Stage 3 as a filter by at most 0.9% at 10 and 60 KB, the budgets it was run at. With $d = 4$ candidates the two rules usually pick the same occupant, so what separates ARGUS-V from ARGUS is the verdict that follows the pick.

Dependence beyond the working model. The working model assumes independent windows; none of the algorithms change under weaker assumptions, only the constant in $\varepsilon(T)$ or its rate. *(a)* If the indicators $X_w^{(u)}$ are negatively associated, the Chernoff–Hoeffding bounds hold in *both* tails for sums of u's own indicators [1]; the override bound is a product of non-increasing indicators, for which negative association gives the same product bound as an inequality. Theorems V.3(1)–(3), V.4(1)–(3) and 1 carry over unchanged; independence across items and of presence from hashing, and the stopping time $T_0(u)$, remain assumed, the post-admission indicators being negatively associated conditional on $T_0(u)$. *(b)* Under k-wise independence the same bounds hold with a looser constant [2] provided $k = \Omega(\ln(1/\delta))$, k-wise independence capping the attainable tail at $e^{-\Omega(k)}$. *(c)* Under m-dependence, splitting the windows into m interleaved i.i.d. subsequences of length T/m and union-bounding over them gives $\varepsilon = \sqrt{m\ln(2m/\delta)/(2T)}$: the effective sample size is T/m and the margin inflates by about $\sqrt{m}$. A bounded-difference martingale gives the same $1/\sqrt{T}$ rate by Azuma's inequality [3], with a mixing time in place of m under mixing conditions [4]. *(d)* Under long-range dependence ($H > 1/2$) the effective sample size grows only as T^{2-2H} and a $1/\sqrt{T}$ margin is over-confident for long histories. We claim no coverage here and test it empirically (§VI-D of the paper).

Persistence over a jumping window. Persistence is defined over the stream so far, as in prior work, or over a *jumping window* of the last L windows: two ARGUS instances alternate with resets every L windows, standard for sliding-window sketches. A fully sliding definition needs per-occupant window histories.

II. PROOFS

This section proves the statements of §V of the paper and those of §I that admit proofs, in the order in which they are used. Two conventions recur. Every argument conditions on the admission window $T_0(u) = s$, which is a stopping time of u's presence process, so the windows after it carry i.i.d. indicators and the bounds obtained, being uniform in s, hold unconditionally. And wherever a statement concerns the stored $\widehat{V}(u)$ rather than the episode's own count $V^{(\tau)}(u)$, it is either stated jointly with the full-tracking event or obtained from Lemma V.1, which bounds the one by the other deterministically.

A. Proof of Lemma V.1

Proof: We argue field by field, distinguishing increments (which can grow a sketch field) from decay and eviction (which can only shrink it).

Bound on $\widehat{V}$. We argue inside u's current tracking episode, of length $\tau = T(u)$, and write $V^{(k)}(u)$ for the number of its

first k windows that contain u; the claim is $\widehat{V}(u) \le V^{(k)}(u)$ for every $k \le \tau$, by induction on k. At the admission that opens the episode $\widehat{V}(u) = 1$ and the admission window contains u, so the claim holds at $k = 1$. For the step, Stage 1 increments $\widehat{V}(u)$ only at an arrival of u found with $\mathrm{status} = 0$ and sets the flag at once, so within the episode at most one increment occurs per window and only in a window containing u; Stage-4 decay decrements $\widehat{V}(u)$, and a replacement or a variant's shortcut ends the episode (an item evicted and re-admitted in the same window opens a *new* episode with $\widehat{V} = 1$, which is why the bound is stated per episode). Hence $\widehat{V}(u)$ grows by at most one from window k to window $k+1$ and only when window $k+1$ contains u, which is the step. Finally the episode's presence windows are a subset of u's, so $V^{(\tau)}(u) \le V_t(u)$.

Bound on $\widehat{C}$. Each Stage-1 match increments $\widehat{C}$, and matches are a subset of u's arrivals, so $\#\{\text{increments}\} \le C_t(u)$. Decay and eviction only decrease $\widehat{C}$.

Bound on $\widehat{\pi}$. The update $\widehat{\pi} \leftarrow \max(\widehat{\pi}, \pi(a))$ over a subset $\mathcal{M}(u) \subseteq \{a : a \in u\}$ of u's arrivals gives $\widehat{\pi}(u) = \max_{a \in \mathcal{M}(u)} \pi(a) \le \max_{a \in u} \pi(a) = \pi_t(u)$. Eviction zeroes $\widehat{\pi}$. ∎

B. Proof of Theorem V.2

Proof: QUERY reports $u \in R_\star$ only if $\widehat{V}(u)/T \ge \alpha$, and $u \in R_{\mathrm{PP}}$ only if additionally $\widehat{\pi}(u) \ge \pi_{\min}$. By Lemma V.1, $V_t(u)/T \ge \widehat{V}(u)/T \ge \alpha$ and $\pi_t(u) \ge \widehat{\pi}(u) \ge \pi_{\min}$; the latter is $\psi_{\mathrm{PP}}(u) = 1$. ∎

C. Proof of Theorem V.3

Proof: Throughout, condition on $T_0(u) = s$: $T_0(u)$ is a stopping time of u's presence process, so the indicators of windows $s+1, s+2, \ldots$ are i.i.d. $\mathrm{Bernoulli}(p(u))$, and since admission is triggered by an arrival of u the window s contains u. Hence $V^{(\tau)}(u) = 1 + S$ with $S \sim \mathrm{Binomial}(\tau - 1, p(u))$, the $\tau - 1$ post-admission trials. All bounds below are uniform in s and therefore hold unconditionally.

Claim (1). By Lemma V.1, $\widehat{V}(u) \le V^{(\tau)}(u) = 1 + S$ deterministically. If $\hat{p}(u) \ge p(u) + \varepsilon(\tau) + 1/\tau$ then $1 + S \ge \tau(p(u) + \varepsilon(\tau)) + 1$, i.e. $S \ge \tau(p(u) + \varepsilon(\tau))$, whence $S/(\tau-1) - p(u) \ge (p(u) + \tau\varepsilon(\tau))/(\tau-1) \ge \tau\varepsilon(\tau)/(\tau-1)$. Hoeffding's upper tail on the $\tau - 1$ post-admission trials bounds the probability of the larger event by $e^{-2(\tau-1)(\tau\varepsilon(\tau)/(\tau-1))^2} = e^{-2\tau^2\varepsilon(\tau)^2/(\tau-1)} \le e^{-2\tau\varepsilon(\tau)^2} = \delta/2$, using $\tau^2/(\tau-1) \ge \tau$. For $\tau \ge 2$ and $\delta \le 2/e$, $2/\tau \le 1 \le \ln(2/\delta)$ gives $1/\tau \le \varepsilon(\tau)$, so $\{\hat{p}(u) \ge p(u) + 2\varepsilon(\tau)\}$ is contained in the event just bounded.

Claim (2). We prove the more general statement $\Pr[\{\hat{p}(u) \le p(u) - t\} \cap \mathcal{F}_\tau(u)] \le e^{-2\tau t^2}$ for every $t > 0$. On $\mathcal{F}_\tau(u)$, $\widehat{V}(u) = 1 + S$, so $\hat{p}(u) \le p(u) - t$ forces $1 + S \le \tau(p(u) - t)$ and hence $S - (\tau - 1)p(u) \le p(u) - \tau t - 1 \le -\tau t$, using $p(u) \le 1$. Hoeffding's lower tail on the $\tau - 1$ post-admission trials bounds the probability of the larger event by $e^{-2(\tau t)^2/(\tau-1)} \le e^{-2\tau t^2}$, again by $\tau^2/(\tau-1) \ge \tau$. Claim (2) is the case $t = \varepsilon(\tau)$, whose bound is $\delta/2$; the case $t = \Delta + \varepsilon(\tau)$ is used in Theorem V.4(2). The statement is joint by necessity: $\mathcal{F}_\tau(u)$ is not a monotone function of u's indicators, since raising $\mathrm{LCB}(u)$ changes which candidate is victimized, so no correlation inequality applies and the conditional form costs a factor $1/\Pr[\mathcal{F}_\tau(u)]$. Neither this argument nor the one for claim (1) uses the binomial law beyond Hoeffding's inequality, which is why item (a) of *Dependence beyond the working model* in §I can replace independence by negative association.

Claim (3). The complement of (1) reads $p(u) > \hat{p}(u) - \varepsilon(\tau) - 1/\tau$ and the complement of (2), in its joint form, reads $p(u) < \hat{p}(u) + \varepsilon(\tau)$; a union bound over the two failure events gives δ. Intersecting with $p(u) \in [0, 1]$ shows that clipping preserves coverage. ∎

D. Proof of Theorem V.4

Proof: (1) Wrong protect. Protect fires iff $\hat{p}(u) - \varepsilon(\tau) \ge \alpha$. Under $p(u) \le \alpha - 1/\tau$ this implies $\hat{p}(u) - p(u) \ge \alpha - p(u) + \varepsilon(\tau) \ge \varepsilon(\tau) + 1/\tau$, whose probability is at most $\delta/2$ by Theorem V.3(1), unconditionally.

(2) Wrong evict via UCB. $\mathrm{UCB}(u) < \alpha$ iff $\hat{p}(u) + \varepsilon(\tau) < \alpha$. Under $p(u) \ge \alpha + \Delta$ this implies $\hat{p}(u) - p(u) < -(\Delta + \varepsilon(\tau))$. The general form of Theorem V.3(2) with $t = \Delta + \varepsilon(\tau)$ bounds this jointly with $\mathcal{F}_\tau(u)$ by $e^{-2\tau(\Delta+\varepsilon(\tau))^2} \le \min\{e^{-2\tau\varepsilon(\tau)^2}, e^{-2\tau\Delta^2}\} = \min\{\delta/2, e^{-2\tau\Delta^2}\}$.

(3) Override. The override fires only if $\widehat{V}(u) = 1$ and $1/\tau < \alpha$, i.e., $\tau > 1/\alpha$. Jointly with $\mathcal{F}_\tau(u)$, $\widehat{V}(u) = 1$ at tracked length τ means $V^{(\tau)}(u) = 1$, and since the admission window contains u this says that u did not appear in the $\tau - 1$ windows following its admission window, an event of probability $(1 - p(u))^{\tau-1}$ under the model. Hence the override fires with probability at most $g_\alpha(\tau) = (1 - p(u))^{\tau-1}\mathbf{1}[\tau > 1/\alpha] \le (1 - \alpha)^{\tau-1}\mathbf{1}[\tau > 1/\alpha]$. The largest value over admissible τ is attained at $\tau = \lfloor 1/\alpha \rfloor + 1$ and equals $(1 - \alpha)^{\lfloor 1/\alpha \rfloor}$. Summing the geometric series from that τ on, $\sum_{\tau > 1/\alpha}(1 - \alpha)^{\tau-1} = (1 - \alpha)^{\lfloor 1/\alpha \rfloor}/\alpha = G_\alpha$. ∎

E. Proof of Theorem 1

Proof: On $\mathcal{F}_T(u)$ with $\hat{\psi}_\star(u) = \psi_\star(u) = 1$ the QUERY gate reduces to $V_t(u)/T \ge \alpha$. Writing $V_t(u) = \sum_{w=1}^{T} X_w^{(u)}$ with $\mathbb{E}[X_w^{(u)}] = p(u) \ge \alpha + \Delta$, a sum over all T windows so that no admission-window bias enters,

$$\begin{aligned}\Pr[\{u \notin R_\star\} \cap \mathcal{F}_T(u)] &\le \Pr\Big[\tfrac{1}{T}\textstyle\sum_w X_w^{(u)} < \alpha\Big] \\ &\le \Pr\Big[\tfrac{1}{T}\textstyle\sum_w X_w^{(u)} - p(u) < -\Delta\Big] \le e^{-2T\Delta^2}\end{aligned}$$

by one-sided Hoeffding; no conditioning on $\mathcal{F}_T(u)$ is used. Corollary 6 follows by solving $e^{-2T\Delta^2} \le \eta$. ∎

F. Proof of Theorem 2

Proof: The deterministic bound is Lemma V.1. For the probabilistic bound, recall that Stage 1 causes u to occupy a single row $i^*(u)$ while tracked. Call an arrival *adverse* for u if it triggers a Stage-4 replacement or decrement of $\widehat{V}(u)$; on $\mathcal{D}(u)$ these are the only events that can drive $\widehat{V}(u) < V_t(u)$. We count only the *first* one: for $u' \ne u$ and a window j, let $E'_{j,u'}$ be the event that the first adverse arrival on u is an

arrival of u' in window j. These events are disjoint and

$$\{\widehat{V}(u) < V_t(u)\} \cap \mathcal{D}(u) \subseteq \bigcup_{j \le T} \bigcup_{u' \neq u} E'_{j,u'}.$$

(Counting every adverse arrival instead would not do: a newcomer arriving k times in one window can decrement $\widehat{V}(u)$ up to k times, whereas Stage 3 bounds only the number of *replacements* per bucket per window.) Just before the first adverse arrival $\mathcal{F}_{\tau_j}(u)$ holds, where $\tau_j = j - T_0(u) + 1$ is u's tracked length in window j, so the joint per-collision bounds of Theorem V.4(2)–(3) apply. By the independence convention of the theorem statement (exact for the arrival and the hash collision under uniform hashing), and using that u being selected as victim has probability at most 1,

$$\Pr[E'_{j,u'}] \le p(u') \cdot \frac{1}{w} \cdot a_{\mathrm{LCB}}(u; \tau_j),$$
$$a_{\mathrm{LCB}}(u;\tau) \le \min\{\delta/2,\, e^{-2\tau\Delta^2}\} + g_\alpha(\tau),$$

the two summands being the UCB path and the override path of Theorem V.4(2)–(3), combined by a union bound. Summing the disjoint events over u' and j,

$$\Pr[\{\widehat{V}(u) < V_t(u)\} \cap \mathcal{D}(u)] \le \frac{\|p\|_1 - p(u)}{w} \sum_{\tau=1}^{T} a_{\mathrm{LCB}}(u;\tau)$$
$$\le \frac{\|p\|_1 - p(u)}{w} \Big(\min\Big\{ \frac{T\delta}{2},\, \sum_{\tau \ge 1} e^{-2\tau\Delta^2} \Big\} + G_\alpha \Big),$$

and $\sum_{\tau \ge 1} e^{-2\tau\Delta^2} = 1/(e^{2\Delta^2} - 1) \le 1/(2\Delta^2)$ gives (1). No appeal to Markov's inequality is needed. ∎

G. Proof of Corollary 3

Proof: Write $\mathcal{N} := \{u \notin R_{\mathrm{PI}}\}$ and split on $\mathcal{F}_T(u)$:

$$\begin{aligned}\Pr[\mathcal{N} \cap \mathcal{D}(u)] &= \Pr[\mathcal{N} \cap \mathcal{F}_T(u) \cap \mathcal{D}(u)] \\ &\quad + \Pr[\mathcal{N} \cap \mathcal{F}_T(u)^c \cap \mathcal{D}(u)] \\ &\le \Pr[V_t(u)/T < \alpha] \\ &\quad + \Pr[\mathcal{F}_T(u)^c \cap \mathcal{D}(u)],\end{aligned}$$

where Theorem 1 bounds the first term by $e^{-2T\Delta^2}$ and Theorem 2 the second by the collision term. Both steps are joint; no conditioning on $\mathcal{F}_T(u)$ is used. ∎

H. Proof of Theorem V.5

Proof: $u \in R_\star$ requires $\hat{\psi}_\star(u) = 1$, so $\psi_\star(u) = 0$ forces $\hat{\psi}_\star(u) \neq \psi_\star(u)$ at query time and $\Pr[u \in R_\star,\ \psi_\star(u) = 0] \le \Pr[\hat{\psi}_\star(u) \neq \psi_\star(u)] \le \bar{q}_\star$. The count part of the guarantee is Theorem V.2. ∎

Measured over the 30 (stream, budget) points of §VI of the paper, precision is at least 0.998 for PP, 0.985 for PH and 0.979 for PS, so the realised predicate-misjudgement rate is at the 10^{-2} level and the F1 gaps reported there are recall gaps. PP is the deterministic case of the three: its priority field is a maximum, which cannot overestimate, so its precision is that of Theorem V.2 read over fingerprints.

I. Proof of Corollary 4

Proof: On $\mathcal{F}_T(u) \cap \mathcal{D}(u)$ with $\psi_\star(u) = 1$ we have $\widehat{C} = C_t$, $\widehat{V} = V_t$ and $\widehat{\pi} = \pi_t$, hence $\hat{\psi}_\star(u) = 1$ (Remark 5), so $\{\hat{\psi}_\star(u) = 0\}$ adds nothing to $\mathcal{F}_T(u)^c$ and, splitting as in Corollary 3, $\Pr[\{u \notin R_\star\} \cap \mathcal{D}(u)] \le e^{-2T\Delta^2} + \Pr[\mathcal{F}_T(u)^c \cap \mathcal{D}(u)]$. For the second term, the variant Stage 4 has one additional adverse path: the shortcut fires when $\hat{\psi}_\star(u) = 0$ and $\mathrm{LCB}(u) < \alpha$, which at tracked length τ has probability at most $q_\star(\tau)$ jointly with $\mathcal{F}_\tau(u)$. Adding $q_\star(\tau)$ to $a_{\mathrm{LCB}}(u;\tau)$ in the proof of Theorem 2 contributes $\sum_{\tau \le T} q_\star(\tau)$ to the sum over τ, which yields the collision term of (3). For ARGUS-PP, $q_{\mathrm{PP}}(\tau) = 0$ and (3) reduces to (2). ∎

J. Proof of Corollary 7

Proof: The two conditions bound the two terms of (3) by $\eta/2$ each, in the order in which (4) states them: the first gives $e^{-2T\Delta^2} \le \eta/2$ and the second makes the collision term at most $\eta/2$. ∎

K. Proof of Proposition V.6

Proof: At $\tau = 1$ the occupant was admitted in the current window, so $\mathrm{status} = 1$ and Stage 3, which both rules share, returns before any verdict: $a_{\mathrm{LCB}}(u;1) = a_V(u;1) = 0$. Let $\tau \ge 2$.

(1) Under ARGUS-LCB an adverse event on u requires u to be the Stage-2 victim (probability at most 1), $\mathrm{status}(u) = 0$, and then either $\mathrm{UCB}(u) < \alpha$ or the override; jointly with $\mathcal{F}_\tau(u)$ these are bounded by Theorem V.4(2)–(3) and a union bound, giving $a_{\mathrm{LCB}}(u;\tau) \le \min\{\delta/2, e^{-2\tau\Delta^2}\} + g_\alpha(\tau)$, a function of τ, δ, Δ and $p(u)$ only and therefore uniform over the other candidates' states. Both summands vanish as $\tau \to \infty$ for $\Delta > 0$ (the first because $e^{-2\tau\Delta^2} \to 0$, the second geometrically). At $\tau = 2$ and $\alpha \le 1/2$ neither path is available: if $\widehat{V}(u) = 1$ then $\mathrm{UCB}(u) = \min\{1, 1/2 + \varepsilon(2)\} \ge 1/2 \ge \alpha$ and the override needs $\tau > 1/\alpha \ge 2$; if $\widehat{V}(u) = 2$ then $\hat{p}(u) = 1$ and $\mathrm{UCB}(u) = 1$. Hence $a_{\mathrm{LCB}}(u;2) = 0$.

(2) Under ARGUS-V the victim is the count-argmin and every victim with $\mathrm{status} = 0$ suffers an adverse event (replacement if $V = 1$, decrement otherwise), so $a_V(u;\tau) = \Pr[u = v_V^*]$ on that event. If u holds the minimum count, it is the argmin up to ties: with uniform tie-breaking $\Pr[u = v_V^*] = \mathbb{E}[1/(1 + N_=(u))] \ge 1/d$, where $N_=(u) \le d - 1$ counts the candidates tied with u; with row-order tie-breaking the same holds after averaging over the uniform placement of u among the d rows. The bound involves neither τ, α, nor $p(u)$. Silence in the $\tau - 1$ windows after admission implies $\widehat{V}(u) = 1$, the smallest count an occupied bucket can hold, and $\mathrm{status}(u) = 0$, for as long as u is tracked; at $\tau = 2$ Stage 3 protects u throughout its admission window, so u is still tracked at the first collision of the next window and the bound $a_V(u;2) \ge (1 - p(u))/d$ is unconditional. For $\tau > 2$ the state $\{\widehat{V}(u) = 1,\ \mathrm{status}(u) = 0,\ u \text{ tracked}\}$ has probability at most $(1 - p(u))^{\tau - 1}$, because ARGUS-V replaces a $V = 1$ occupant at any collision it loses; the lower bound $1/d$ is therefore stated conditionally on that state. ∎

L. Proof of Corollary 8

Proof: (1) Since every other candidate has count strictly greater than V, u and u' are the only count-minima; uniform tie-breaking selects u with probability $1/2$, and ARGUS-V penalizes its victim whenever Stage 3 does not protect it, which the hypothesis $\mathrm{status}(u) = \mathrm{status}(u') = 0$ rules out.

(2) Since every other candidate has a larger unclipped score, the ARGUS-LCB victim is u or u', and with status 0 it reaches Stage 4. If the victim is u, Stage 4 penalizes it only in the evict branch, i.e., if $V/T_1 + \varepsilon(T_1) < \alpha$ or ($V = 1$ and $V/T_1 < \alpha$, which needs $T_1 > 1/\alpha$); both require $V/T_1 < \alpha$, so an item with $V/T_1 \geq \alpha$ is protected or deferred. If the victim is u' and $V \geq 2$, only the UCB path is available, so it is penalized iff $V/T_2 + \varepsilon(T_2) < \alpha$; if $V = 1$, either path fires iff $1/T_2 < \alpha$, since $1/T_2 + \varepsilon(T_2) < \alpha$ implies $1/T_2 < \alpha$ and, for $T_2 > 1/\alpha$, the override fires on an occupant whose single observed window is now a vanishing fraction of its history. ∎

III. Comparison with Full Keys

The identifier width is an accounting choice, so this control removes it. Figure 10 of the paper repeats the comparison of §VI-B of the paper with the full key stored in every bucket, which leaves ARGUS and the count-based sketches at the same bucket width. The measurement covers the five real streams over five seeds; the figure plots the three backbone ones. ARGUS is first by 9.5% on average, 14.9% at 10 KB and up to 22.6%, and never by less than 2.4%. The narrowest margins on average are on WebDocs (the single narrowest cell being Campus at 60 KB), where the 784 buckets available at 10 KB face 4,824 persistent items and every method fills its table with true positives.

IV. The Bucket Width Control

Throughout §VI of the paper every method is charged the width its own bucket fields need, so the methods do not hold the same number of buckets, exactly as they would not in a deployment. This control removes that difference, leaving the decision rule. Figure 1 charges the count-based sketches ARGUS's bucket width instead of the wider one their fields need, which spend a field of their own on the status flag that ARGUS packs beside its first-tracked window, which gives each of them 12.5% more buckets than its layout admits, so that they and ARGUS hold exactly the same number; the other baselines keep their published layouts. ARGUS still leads, by 6.6% on average, 15.3% at 10 KB and up to 22.2%, against 10.0%, 22.3% and 27.3% when each method is charged its own width.

V. Additional Figures and Settings

This section collects the settings behind §VI of the paper and the figures it has no room to print. The numbers a figure supports are in the paper and are not repeated here, except where a figure needs one to be read. **The hardware design.** Replacing the status bit by the window index of the last arrival, as §VI-G of the paper describes, widens the bucket; the validation there runs on $4{,}096$ buckets.

Hardware resources (Table IV of the paper). The place-and-route result behind §VI-G of the paper: resources and achieved clock at three sketch sizes on a Kintex UltraScale+ XCKU5P, out of context. The logic is almost fixed across the three, under 2.2% of the device and 23 DSPs at every size, so what scales with the sketch is block RAM and what falls with it is the clock. The critical path runs from a block-RAM read through the table lookup, multiplier and four-way comparison back into a write, and lengthens with the address width.

Synthetic stream parameters. The ten generated streams of §VI-D of the paper are: Zipf popularity with $s \in \{0.5, 0.75, 1.0, 1.25\}$, at 20 M arrivals over 700 K items; Markov-modulated presence with $\phi \in \{0, 0.6, 0.9\}$; long-range-dependent on/off presence with Pareto period lengths, $a \in \{1.5, 1.2\}$, giving $H \in \{0.75, 0.9\}$; and deterministic on/off cycles of length 100.

Coverage snapshots. The snapshots behind Figure 5 are taken every 50 windows on the full-key build. They also give the tenure distribution quoted in §III-A of the paper: at 60 KB 12.5% of occupant snapshots have been tracked for fewer than 16 windows and 7.6% for fewer than four, against 1.2% below 16 at 10 KB, where the table is close to frozen and its occupants are correspondingly long-lived.

Baseline settings. Three baselines split the budget between two structures. Waving-Sketch gives half to its Bloom filter of per-window presence and half to the heavy part. TinyLFU gives 25% to its count-min filter of per-window presence, whose counters are halved every 128 windows, and the rest to the bucket array. The PP composites give 35% to the priority layer and 65% to the persistence sketch beside it. Every other baseline uses the layout and constants of its paper, at the $d = 4$ rows of §VI-A of the paper where it takes a row count, and §VI-D of the paper searches both.

Reading Figure 3 of the paper. In the right panel the measured miss rates are 0.31, 0.05 and 0.02 at $\Delta = 0.0025$, 0.0175 and 0.025, against bounds of 0.98, 0.38 and 0.14, and zero from $\Delta = 0.03$ on. At 1 MB no miss is a collision, so the Hoeffding term is tested alone, and the curve is not monotone because the bins nearest α are the smallest, 22 to 24 items against 122 in the farthest. Theorem V.4(1) covers the longer-tracked of the occupants in the left panel. The 100 KB series of the left panel rests on about 2,400 verdicts against about 220,000 at 30 KB, which is why its points move more between settings.

Synthetic streams (Figure 7 of the paper). This figure covers ten generated streams. Each varies one property of the stream while holding the others fixed, and each is run against the same six baselines as Figure 4 of the paper and over the same six budgets. The four Zipf panels vary popularity skew, the three Markov panels the autocorrelation of an item's presence across consecutive windows, the two long-range-dependent panels the Hurst exponent of an on/off process with Pareto period lengths, and the last panel is a deterministic cycle. Only the Zipf panels and the Markov $\phi = 0$ panel satisfy the working model of §III-A of the paper; the remaining Markov panels, the two long-range-dependent ones and the cycle violate its

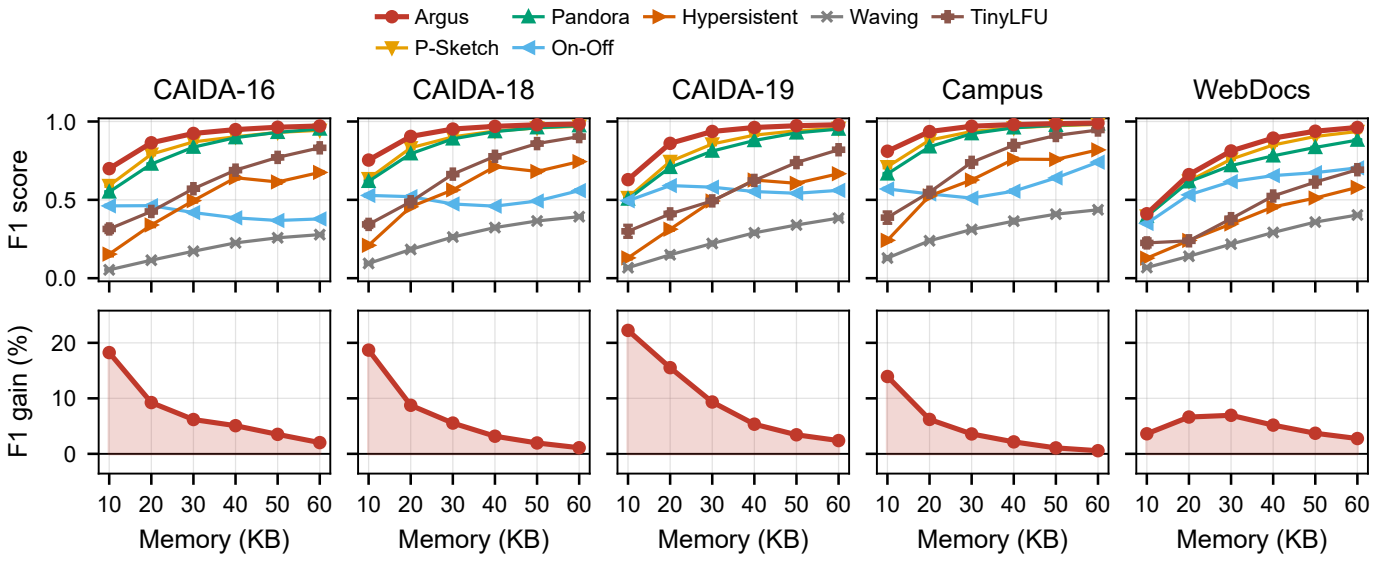


Fig. 1. F1 with the count-based sketches handed ARGUS's bucket width, so that they and ARGUS hold the same number of buckets while the other baselines keep their published layouts, and the gain over the best baseline at each point (five real streams, five seeds). Compare Figure 4 of the paper, where each method is charged the width its own fields need.

independence assumption by construction, and are the point of the figure. Read each group in order, the Markov group wrapping to the second row: the question is not whether ARGUS leads, which it does throughout, but whether its margin survives as the violation deepens.

Threshold and window grid (Figure 8 of the paper). The paper fixes $T = 1{,}600$ windows and $\alpha = 0.3$; this grid varies both, five window counts by six thresholds, each cell scored against its own ground truth. One seed per cell, so individual cells are noisy. The figure pools the 30 cells into one F1 distribution per method, and therefore reports how each method holds up across the grid rather than any trend within it. Averaged over the window counts, ARGUS's margin over the best baseline is widest at the low thresholds and narrows to 4.2% at $\alpha = 0.7$, because a low threshold admits a large qualifying set whose members all compete for the same buckets, and a high one admits a set small enough for every method to hold.

Where updates end (Figure 2). Every update is classified by the stage that disposes of it. The two panels give the tightest and the loosest budget, so they show the same rule, on each stream, under opposite contention. The breakdown is measured on the full-key build, which is the one whose instrumentation records the per-stage counters. That layout uses a wider bucket, so it holds fewer buckets at the same budget and the contention it reports is the higher of the two. Of the collisions at 10 KB, 27–45% end in protect, 9.7–56% in defer and 17–45% in Stage-3 drops, while only 0.02–0.5% change the table, so under contention the table is nearly frozen around the occupants the lower bound confirms. At 60 KB deferrals still dominate, with decay at 2.3–11.2% of collisions and replacements at 0.3–5.9%. Of the replacements, 68–90% come from the $V{=}1$ override, by design: at $V = 1$ the interval spans almost the whole unit range, the case §III-D of the paper introduces the override for and Theorem V.4(3) bounds. In the other direction, 0.15 to 5.1% of the items ARGUS evicts are persistent on the four network streams; the exceptions are all on WebDocs, where capacity rather than the rule binds and the share reaches 20% at 10 KB before falling back inside the band by 50 KB.

The third outcome and the margin's width. §VI-E of the paper separates removing the third outcome from changing the width of the margin that allows it. Removing it, by setting the margin to zero, leaves a two-way rule 3.4% behind ARGUS at 60 KB and 1.0% behind at 30 KB but 4.0% ahead at 10 KB: there the margin is wide and almost nothing reaches the table at all, as the breakdown above shows, and a table this contended can less afford to wait for evidence. Deferring is what the margin buys, and it pays once collisions stop saturating the table; removing it also forfeits Theorem V.4, which is stated for the three-way rule. Changing the width instead moves little: doubling or halving it costs at most 0.6% of F1 at 60 KB, no schedule tried moves F1 by more than 0.7% there, and the Wilson, Clopper–Pearson and empirical-Bernstein intervals stay within 0.2% of it there. At 60 KB the only schedule ahead of $\varepsilon(T)$ is a constant 0.1, by 0.5%, and it is ahead at every budget by 0.6% on average; averaged over the budgets seven of the ten tried are ahead, by as much as 1.0% for a constant 0.05, which nonetheless reverses to -0.4% at 60 KB. Both constants are the opposite schedule to $\varepsilon(T)$, narrow where it is wide and wide where it is narrow, so they commit early on newcomers and wait on the established, against the tenure distribution of §III-A of the paper, and neither carries a miscoverage level, so Theorem V.4 says nothing about them. Clopper–Pearson was run at 10, 30 and 60 KB only.

Error of the reported persistence (Figure 3). Per budget and stream, on a log scale because the plotted values span

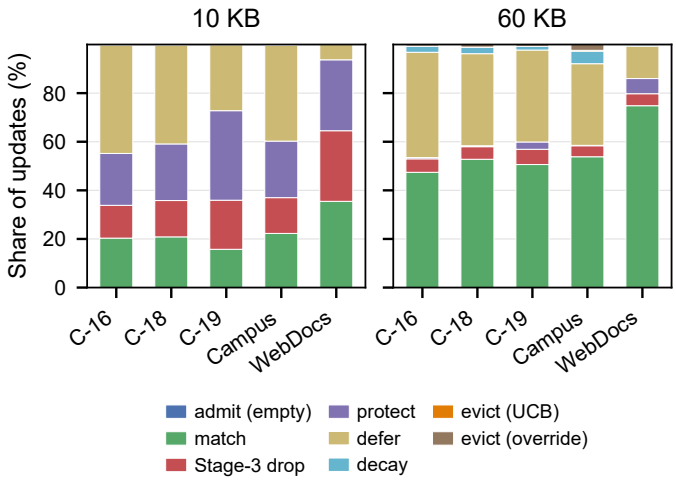


Fig. 2. Outcome of every update, by stage, for ARGUS at 10 and 60 KB. Two bands are too small to see at this size: admitting into an empty bucket accounts for at most 0.05% of updates, and eviction on the upper bound for at most 0.36%.

three orders of magnitude. The caveat in the caption is the one to keep in mind throughout: the metric is averaged over true positives only, so a method that reports fewer items is scored on an easier subset, and the additional true positives ARGUS finds are the borderline ones whose estimates are noisiest. The figure is therefore a lower bound on ARGUS's advantage, not an upper one. Figure 5 of the paper summarises the same data as one distribution per method.

The remaining figures carry the measurements §VI of the paper reports without room to plot: the parameter sweeps and the coverage of the interval.

Parameters (Figure 4). Each parameter is swept with the other held at its default, $\delta = 0.05$ and $d = 4$. The left panel is the one that matters for the design claim. δ enters only through $\sqrt{\ln(2/\delta)}$, so the twentyfold range swept rescales the margin by about $1.5\times$, and the curves are correspondingly flat. The right panel is an ordinary capacity trade, more rows against fewer buckets each, and it is read for the knee rather than for a maximum.

Coverage (Figure 5). The model rate $p(u)$ of §V of the paper is not observable on real traffic, so the interval is tested against two observable substitutes, by tracked length and at both budgets. Left: the occupant's rate over its own tracked windows. Under full tracking the interval is centred on exactly this quantity and covers it identically, so a miss means only that decay has removed more than $\varepsilon(T(u))\,T(u)$ windows of evidence, and the lower bound never fails; the panel is a test of the sketch's fidelity, not of the statistics. Right: the rate over the whole stream, which no per-occupant estimator can track once an item's activity is confined to a phase. Neither panel can falsify Theorem V.3(3), which is stated jointly with full tracking and about the model rate; they report only how often the interval brackets the two observable readings. The 60 KB shortfall against the tracked-window rate is confined to occupants tracked beyond 64 windows, where $\varepsilon(T)$ is narrow enough for accumulated decay to show. Against the tracked-window rate, two-sided coverage per stream, over all tracked lengths, is 0.999 to 1.000 at 10 KB and 0.946 to 0.994 at 60 KB, and the lower bound never fails, since decay only lowers $\hat{p}$. Against the whole-stream rate it is 0.67–0.96, with 75 to 91% of the failures on the lower bound, which is the direction the design tolerates best: an over-confident lower bound delays an eviction, it does not cause one.

Throughput (Figure 12 of the paper). Two measurements separate the rule from the layout it runs on. An unpacked build, the same rule and bucket count but a byte-addressed struct in place of the packed bucket and victims ranked by count, runs at 13.2 Mops against ARGUS's 17.0, so most of the margin over the count-based sketches is the memory read per probe rather than the branch taken at a collision. The Wilson and empirical-Bernstein margins need a square root per candidate instead of a table lookup, which costs 42 to 48% of throughput for an F1 difference of at most 1.5% at any budget.

Tuned baselines (Figure 9 of the paper). Every baseline is searched per stream and per budget, one axis at a time, over its rows where it has them and over its main constant, and plotted at its best configuration. The grids are rows $d \in \{2, 3, 4, 6\}$ for P-Sketch, Pandora and TinyLFU, the others not being row-parameterised; P-Sketch's replacement factor $\{6, 12, 18, 36\}$, Pandora's inactivity constant $\{10, 25, 50, 100\}$, On-Off's slots per bucket $\{2, 4, 8\}$, Waving's heavy-part size $\{8, 16, 32\}$, Hypersistent's hot-part share $\{0.3, 0.4, 0.5, 0.6\}$, TinyLFU's table share $\{0.6, 0.75, 0.9\}$. ARGUS is plotted at the single default it uses everywhere else in the paper. The search uses one seed per configuration against ARGUS's five, so the baselines are if anything flattered by selection on noise, and the narrowest cells should be read as ties. This is the control that answers whether the comparison in §VI-B of the paper turns on configuration: the per-cell improvement from tuning and the margin that remains are both in §VI-D of the paper. Such a per-cell search is not available to a deployment, which is why it is a control and not the headline.

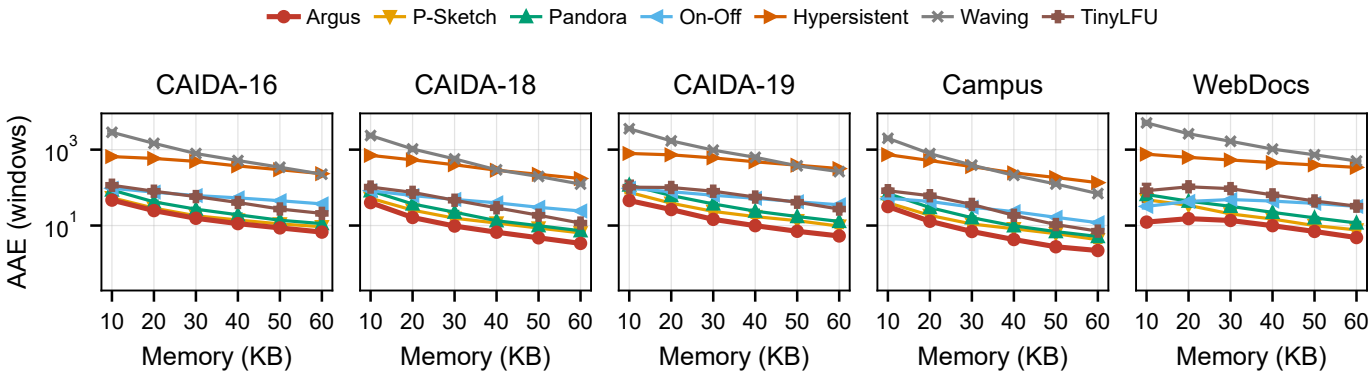


Fig. 3. Average absolute error of the reported persistence for the items each method finds (log scale, five seeds). The metric averages over true positives only, so a method that finds fewer of them is scored on an easier subset.

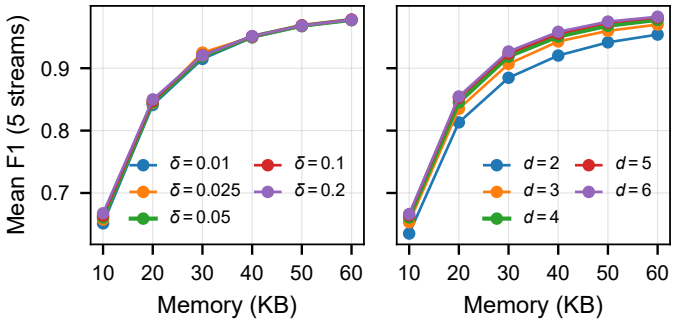


Fig. 4. Mean F1 over the five streams under varying $\delta \in \{0.01, 0.025, 0.05, 0.1, 0.2\}$ (left) and $d \in \{2, 3, 4, 5, 6\}$ (right), each swept with the other at its default. The heavy-task sweep of §VI-H of the paper uses the coarser $\delta \in \{0.01, 0.05, 0.1, 0.2\}$.

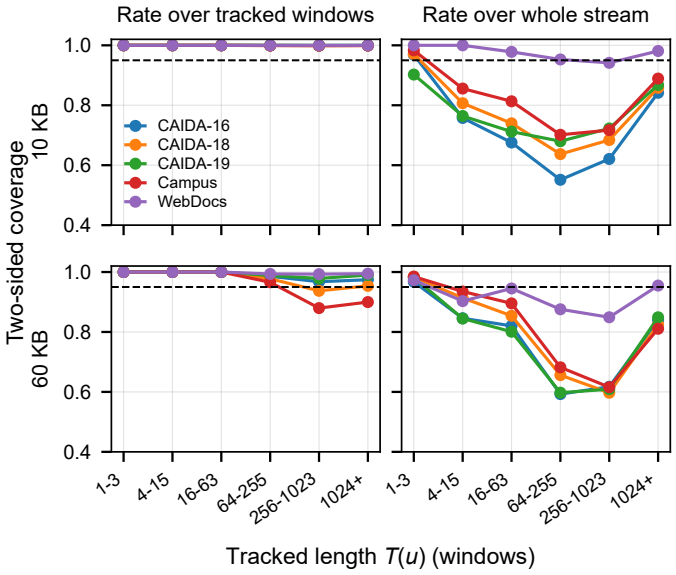


Fig. 5. Two-sided coverage of $[\mathrm{LCB}, \mathrm{UCB}]$ on the real streams at 10 and 60 KB, by tracked length, against the occupant's rate over its tracked windows (left) and over the whole stream (right). The dashed line is the nominal $1 - \delta$. Theorem V.3(3) states it for $[\mathrm{LCB} - 1/T(u), \mathrm{UCB}]$ and for the model rate, jointly with full tracking; the plotted interval omits the $1/T(u)$ admission-bias term and neither target is the model rate, so the line is a reference, not a threshold either panel must clear.

References

[1] D. Dubhashi and D. Ranjan, "Balls and bins: A study in negative dependence," *Random Structures & Algorithms*, vol. 13, no. 2, pp. 99–124, 1998.

[2] J. P. Schmidt, A. Siegel, and A. Srinivasan, "Chernoff–Hoeffding bounds for applications with limited independence," *SIAM Journal on Discrete Mathematics*, vol. 8, no. 2, pp. 223–250, 1995.

[3] K. Azuma, "Weighted sums of certain dependent random variables," *Tohoku Mathematical Journal*, vol. 19, no. 3, pp. 357–367, 1967.

[4] B. Yu, "Rates of convergence for empirical processes of stationary mixing sequences," *The Annals of Probability*, vol. 22, no. 1, pp. 94–116, 1994.